

Turtles are territorial animals, meaning they prefer to be alone in their respective environments. When paired together, turtles can be extremely aggressive towards one another, providing a risk to each of their safety. Our current Tank houses two turtles; Bella, the larger female that has an increased aggression with Zark the smaller turtle that is less aggressive. In the turtle enclosure containing Bella and Zark there lacks a way for the smaller turtle Zark to be able to seek space from Bella when aggression arises.

The turtles experience fights often causing a potential hazard to the physical health of both turtles. Additionally, increasing the amount of time the turtles are independent from each other could potentially boost the quality of the turtle's life within the tank. Making a smaller area where only Zark could fit, can reduce the amount of time the turtles see each other significantly, and lower the amount of risk involved with physical contact between the two turtles.

Bella is the larger turtle in this case meaning that she already has some established dominance over Zark within the enclosure which can be dangerous for Zark if a significant act of aggression was to occur. With turtles' size is an important trait in regard to dominance within an enclosure. In peer reviewed article research displayed the more turtles that were together within a group carried a greater chance of aggressive behavioral patterns, especially with females. (e.g, Schinader, 2010)

Turtles can be very aggressive at times when competing for resources such as food especially. (Polo-Cavia et al.,2011) Aggressive moments between the two turtles are more likely to occur whilst being fed, adding the potential risk of injury to either of the turtles.

Another problem involves Bella consuming more food than Zark at times due to the advantages Bella possesses which could potentially lead to nutritional imbalances for Zark.

Acts of aggression between Bella and Zark are uncommon but not zero; any small bout between them could be the difference between them living or potentially experiencing a life-threatening injury.

In private pet ownership of pet turtles, owners observe and highlight the dominating behavior turtles can exhibit when co-existing alongside each other. Turtles in these personal environments also face competition in regard to being fed, as well as competition being common especially when the Tank size is compact. (e.g, Redearslider , 2011.)

In the care and maintenance of captive sea turtles in captive care facilitates, the turtles in these environments also deal with problems with aggression towards one another. It is emphasized that these problems occur specifically when feeding similarly to Bella and Zark alongside private pet ownership.

Our stakeholders are the Turtles and caretakers. Our secondary stakeholders are our field experts

Dr. Kimble and Mr. Jorge

Concern of the primary stakeholder groups

The enclosure would also support the caretakers by making the process of delivering food easier. When feeding the turtles, we can guarantee that at least hunger needs are met without one of the turtles consuming more food than the other. An enclosure would also allow us to potentially add decorations to the roof area to create a vibrant aquatic environment for the turtles which in turn could also reduce stress levels for the turtles. Alternatively, An enclosure could potentially cause the tank to feel smaller which would not favor Bella causing her quality of life within the tank to decrease due to turtles preferring to be in a more spacious environment.

The enclosure would be about 8.3 inches tall by 16 inches deep by 12.5 inches wide, with an opening in the center about 8 in long x 6 in wide to allow a personalized space for only zark to personally fit into. The enclosures dimensions will be approximately around $\frac{1}{4}$ to $\frac{1}{3}$ of the enclosures original dimensions to allow Bella enough room to be comfortable on her own.

Sources:

<https://www.fws.gov/sites/default/files/documents/seaturtle-standard-conditions-for-care-2019.pdf>

<https://link.springer.com/article/10.1007/s10530-010-9897-2>

Same as

<https://doi.org/10.1007/s10530-010-9897-2>

<https://www.scielo.br/j/aa/a/CPL4XjrXkt3FST9d7cTjgvB/?lang=en>

<https://www.redearslider.com/cohabitation.html>

Schneider, L., Ferrara, C., & Vogt, R. C.. (2010). Description of behavioral patterns of *Podocnemis erythrocephala* (Spix, 1824) (Testudines: Podocnemididae) (Red-headed river turtle) in captivity, Manaus, Amazonas, Brazil. *Acta Amazonica*, 40(4), 763–770. <https://doi.org/10.1590/S0044-59672010000400018>

Polo-Cavia, N., López, P. & Martín, J. Aggressive interactions during feeding between native and invasive freshwater turtles. *Biol Invasions* 13, 1387–1396 (2011).